Sprint 6 — Plan

Window: Monday 19 October – Sunday 25 October 2026
Freeze: Friday 23 · **Verify on production:** Saturday 24 · **Report:** Sunday 25
Written against: `main` @ `77c6185` · from `roadmap.md` and the sprint 5 results

Sprint goal: flickpond.com runs on AWS, and the editor grows four new tools — rotate and speed, colour, filter presets, and a preview on the seek bar — each built end to end by one person. The first upload of the morning no longer waits minutes for a worker, and a long video's MP4 is encoded in parallel across workers.

One member, one feature, end to end. Weekends count as working days. **Read `known-traps.md` before writing code.**

0. Who has what

Track	Owner	This sprint	Main deliverable
A	Ibrahim Mammadov (@1brahim74)	Processing, AWS, the editor's plug-in seam	The cutover; the seam (Monday); chunked MP4 encoding, phase 1; warm pool and a worker role of its own
B	Zhang Jizhang (@zhanj384)	**Rotate, flip and speed**	Three new edit operations: rotate 90/180/270, mirror, 0.5×–2× with the audio pitch kept
C	Yang Dongwei (@ttydw-ch)	**Colour adjustments**	Brightness, contrast, saturation and gamma sliders, previewed live in the browser, applied by FFmpeg
D	Jiang Yibai (@JiangYibai666)	**Seek-bar preview** and infrastructure	Hovering the seek bar shows that moment's frame; CloudFront on our own domain; alerts that reach someone
E	Lu Jingxing (@Dilute-l)	**Filter presets**	One-click looks (B&W, warm, cool, vintage, vivid) previewed live, applied by FFmpeg

Each feature is yours from the FFmpeg filter to the button. It lives in its own files, plugs into the editor through the seam in 2.1, and cannot block anyone else's — or be blocked by anyone else's after Monday.

1. Decisions already made

- **The cutover happens first, Tuesday evening.** Everything it needs was proven on staging in sprint 5: RDS and ElastiCache private and TLS-only, CloudFront serving segments, spot workers scaling from zero, a 4K HDR upload laddered in 13 min 26 s. The DNS TTL is already 10 s. A and D.
- **New edit operations plug into one seam** (2.1). A builds it on Monday with a template operation and its tests; after that, adding an operation touches no one else's code beyond a one-line registration.

- **A fixed order, whatever the request lists:** `clip → crop → rotate/flip → scale → colour → preset → speed → convert`. Crop stays in the original's pixels because it runs before rotation. Colour and presets run after the HDR tone-map, so they always see standard 8-bit video.
- **The live preview is the browser's, the result is FFmpeg's.** While a slider moves, a CSS filter (C) or a WebGL shader (E) changes the picture on the page; nothing goes to the server until **Process**. How closely the preview matches the result is part of each feature's acceptance.
- **Chunked encoding is two sprints.** Phase 1 (this sprint): the MP4 only, split into keyframe-aligned pieces of about 10 s, encoded on the existing worker pool, joined without re-encoding; behind `WORKER_CHUNKED_MP4`, off by default. Phase 2 (sprint 7): the ladder.
- **A warm pool of one stopped worker** (≈ $4 a month for its disk) brings the overnight first-job wait from 3 min 30 s toward a minute.
- **Workers get their own IAM role.** Today they share the core's and can read secrets they never use, such as the CloudFront signing key.

2. The contracts — A posts 2.1 on Monday morning

2.1 The edit operation seam (A builds it, everyone uses it)

Each operation is one module under `app/worker/edit_ops/` and one schema entry, registered in one line:

Part	Where	What it holds
Parameters	`app/schemas/edit.py`	A strict Pydantic model, added to the `EditOperation` union
Rule	`app/services/media_rules.py`	What the API refuses before queueing (422), and the worker repeats against the real file
Filter	`app/worker/edit_ops/<name>.py`	`build(params, frame) -> (video_filters, audio_filters, frame_after)`
Panel	`frontend/<name>.js` + `.css`	`window.mount<Name>(video, onChange)`, the same contract as the crop box and the clip scrubber (`EDITOR-COMPONENTS.md`)

`frame_after` is the frame size the operation leaves behind, so a scale after a 90° rotation is judged against the rotated frame. A missing panel disables its section with "Coming in the next update", as before.

The parameters, agreed now so nobody waits on anybody:

```
{ "operation": "rotate",  "params": { "degrees": 90 } }              // 90 | 180 | 270
{ "operation": "flip",    "params": { "axis": "horizontal" } }       // horizontal | vertical
{ "operation": "speed",   "params": { "factor": 1.5 } }              // 0.5 – 2.0, steps of 0.25
{ "operation": "adjust",  "params": { "brightness": 0.1, "contrast": 1.2,
                                      "saturation": 1.1, "gamma": 1.0 } }
{ "operation": "preset",  "params": { "name": "warm" } }             // bw | warm | cool | vintage | vivid
```

`adjust` ranges: brightness −0.3…0.3, contrast 0.5…2, saturation 0…3, gamma 0.5…2. Every field is optional; leaving one out leaves that value alone. `speed` with `clip`: the clip is cut on the original's timeline first.

2.2 The seek-bar preview (D builds it, A adds the hook)

On Monday A adds an empty `after_ladder` hook to the ladder job. D's module fills it: one frame every 5 s, 160 px wide, tiled into `outputs/<job>/preview/sprite.jpg`, with a WebVTT index beside it. The API adds `preview_vtt_url` (absent until the sprite exists) to the job shape. The player turns on Plyr's own `previewThumbnails` when the field is there. A failed preview never fails the job.

2.3 Chunked MP4 (A builds it, nobody else depends on it)

The job row does not change. With `WORKER_CHUNKED_MP4=true`, a video longer than 60 s is split into chunks, each queued as its own job on `video_jobs-chunks`, and joined by the worker that finishes the last one. `status` and `output_key` are still written only through `app/repositories/jobs.py`. With the flag off, nothing changes.

3. How work flows this week

1. **Contracts by Monday noon.** A posts 2.1–2.3 in the group. From then on you build against the document, not against someone's branch.
2. **Stand-ins, not waiting.** If your feature needs something not yet merged, write a small default behind a flag or a missing-component note, and say so in the PR. The real piece replaces it when it lands.
3. **One PR a day, small.** Open or update your PR by the end of each day. Branch names: `<track>-sprint6-<topic>`, e.g. `c-sprint6-colour`.
4. **Review within 24 hours.** Ibrahim reviews and merges in the evening. A's own PRs have a named reviewer per item (below); after 24 hours any teammate may approve.
5. **Check-in Wednesday evening.** Each track answers in the group: a draft PR link, and "on track" or "need help".
6. **Take-over Thursday morning.** A feature with no draft PR and no answer by then moves: a free teammate or A picks it up, or it moves to sprint 7. It never holds up anyone else's merge. The report records who started it, who finished it and why.
7. **Verify on staging, then production.** Every feature is checked by its owner on https://47-128-86-77.sslip.io before the freeze, and on flickpond.com on Saturday.

A's PR	Reviewer
The seam (2.1), the preview hook (2.2)	C
Chunked MP4 (2.3)	B
Warm pool, worker role	D

4. Schedule

	Mon 19	Tue 20	Wed 21	Thu 22	**Fri 23 · freeze**	**Sat 24 · verify**	Sun 25
A Ibrahim	Sprint 5 report. **Post 2.1–2.3 by noon.** The seam: registry, fixed order, `frame_after`, a template operation with tests; the empty `after_ladder` hook.	Cutover rehearsal on staging in the morning. **Cutover in the evening** with D: stop writes, `pg_dump` → RDS, MinIO → S3, the A record at Spaceship.	Warm pool; the worker role. Chunked MP4: split on keyframes, chunk jobs.	Chunked MP4: join without re-encoding, the last chunk triggers it; a stopped chunk worker gives its chunk back.	PR complete. A 10-minute video, chunked vs. not, timed on 4 spot workers.	Verify on production; flag stays off unless the timing and the seams pass.	Sprint 6 report
B Zhang	Read 2.1. `rotate` and `flip`: schemas, rules (a 90° rotation swaps width and height).	`rotate` / `flip` filters on the template; tests including rotate + crop + scale.	`speed`: `setpts` and `atempo` (chained for 0.5×), with clip. **Check-in.**	Panel: rotate buttons with a CSS preview, mirror toggles, speed dropdown.	PR complete; verified on staging	Verify on production	Slack
C Yang	Read 2.1. `adjust`: schema, ranges, rules.	`eq` filter on the template, after the tone-map; tests.	Panel: four sliders with a live CSS-filter preview; Reset. **Check-in.**	Match the preview to FFmpeg: compare stills at five settings, tune the mapping.	PR complete; verified on staging	Verify on production	Slack
D Yibai	Read 2.2. Production apply of the worker group and CloudFront, with A.	**Cutover** with A: workers live; arm the health alarm.	Seek-bar sprite and WebVTT in the hook. **Check-in.**	`preview_vtt_url` in the API; Plyr `previewThumbnails`. CloudFront on `cdn.flickpond.com` (ACM in us-east-1, TLS 1.2 minimum).	PR complete. Alerts to an SNS email topic: queue metrics missing, health, budget at $80.	Verify on production. Spot interruption test (AWS FIS) with A.	Write-up: the first week's real AWS cost
E Lu	Read 2.1. `preset`: schema; the five looks as FFmpeg chains (`eq`, `colorbalance`, `curves`).	Filters on the template; tests.	Panel: preset tiles with a live WebGL preview on the video. **Check-in.**	Match the preview to FFmpeg, as C does; one Playwright smoke test of the editor in a real browser.	PR complete; verified on staging	Verify on production	Slack

Checkpoints — things you can check, not statuses you post

By the end of	This is true, or we have a problem
Mon 19	2.1–2.3 posted. The template operation merged, with tests. The sprint 5 report merged.
Tue 20	flickpond.com is served from AWS. A 4K upload plays in 4K. Alibaba is untouched, kept for a week as the rollback.
Wed 21	Every track has a draft PR (the check-in). The first job after an idle hour starts in under 1.5 min.
Thu 22	Each new operation works through the API on staging, without its panel if need be. Take-overs, if any, are decided.
Fri 23	**Freeze:** every track's PR is complete and verified on staging. A chunked 10-minute MP4 has no visible seams.

By the end of	This is true, or we have a problem
Sat 24	On flickpond.com: rotate + speed, colour, a preset and the seek-bar preview each work on one real upload; a spot worker interrupted mid-job gives the job back; alerts reach the email topic.

5. Carried over, folded in above

- **The production cutover** — sprint 5's Saturday, not yet run — A and D, Tuesday.
- **The sprint 5 report** — A, Monday.
- **The spot interruption test with AWS FIS** — D and A, Saturday.
- **CloudFront on our own domain** — today's `*.cloudfront.net` certificate allows TLS 1.0 whatever is asked (#88) — D.
- **Alerts that reach someone** — the queue-metrics alarm has no action — D.
- **A worker role of its own** — found in the workers review — A.
- **Browser tests in CI** — three bugs in sprint 5 passed 245 jsdom tests and only showed in Chromium (#86, #89) — E starts with one smoke test.

6. Not this sprint

- **Admin page with real roles and rights** — sprint 7, with B (accounts).
- **Keep HDR** — an HDR ladder beside the SDR one — sprint 7, with C.
- **Chunked encoding, phase 2: the ladder** — sprint 7, A.
- **Encode by popularity, AV1** — after chunked encoding.
- **Deploys from CI** — after the cutover has run for a week.

7. Rules that have not changed

1. `main` requires a pull request and a review.
2. Ask before editing a file another track owns, and say so in the PR. The one-line registrations in 2.1 are agreed here and need no asking.
3. Every file under `tests/integration/` needs the `RUN_POSTGRES_TESTS` guard.
4. The worker is the sole writer to `status`, `output_key`, `error` and `updated_at` after the API's insert, through `app/repositories/jobs.py` only.
5. Never commit credentials. Check `git diff --cached --name-only` first. Secrets live in SSM Parameter Store.

6. Anything that opens a datastore to the network is reviewed before `apply`, by someone other than whoever runs it.
7. After the cutover, `aws-cutover.md` is the deploy procedure, and CLAUDE.md's deployment section is rewritten to match.
